MOHAMMAD SHIHAB HOSSAIN

Computer Science & Engineering Undergraduate | Data Science

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PROFESSIONAL SUMMARY

Computer Science undergraduate and Data Science researcher specializing in applied machine learning, AI orchestration, and scalable software engineering. Adept at bridging the gap between theoretical research and production-ready architectures. Proven ability to design end-to-end intelligent systems, optimize inference pipelines for high-performance execution, and architect robust backend platforms to solve complex computational challenges.

EDUCATION

American International University-Bangladesh (AIUB) *Expected Dec 2028*

B.Sc. in Computer Science and Engineering | Major Focus: Data Science | **CGPA: loading/4.00**

TECHNICAL SKILLS

Programming: Python, Java, C++, JavaScript/TypeScript, Rust, C#, PHP

AI & Data: Machine Learning, Deep Learning, NLP, LLM Integration, LLM Harness Engineering, Agent Engineering, Model Optimization

Software & Tools: PyTorch, Hugging Face Transformers, scikit-learn, ONNX Runtime, pandas, Docker, Git/GitHub Actions, React, Next.js, Fastify, Node.js, PostgreSQL, MySQL

SELECTED PROJECTS

GridWise LLM | Campus Energy Optimization API | Lead Developer

- Developed a TypeScript/Fastify API that interprets campus operator notes with Mistral, validates directives with Zod guardrails, and falls back to deterministic parsing when needed.
- Implemented a linear-programming optimizer and schedule validator for 24-hour battery/grid plans; passed all 10 published BUP CSE Fest 2026 sample scenarios in a live-endpoint test.

ExoDiscover AI | Lead Developer NASA Space Apps Challenge 2025

- Trained a TabNet classifier on NASA exoplanet-candidate data and exported the model to ONNX for browser-based inference.
- Built a React/TypeScript interface for interactive candidate predictions, probability display, and model feature-importance visualization.

Whisper Flow Linux | Offline Speech-to-Text

- Built a native Rust Linux dictation tool integrating whisper.cpp, microphone capture, voice-activity detection, model management, and clipboard-based text insertion.
- Added lazy model loading and idle unloading; repository benchmarks document 7 MB idle RSS and 136 MB peak RSS with the default quantized model.

House Rental Management Platform | Lead Developer

- Led the PHP MVC architecture, MySQL schema, authentication, tenant portal, and public marketplace for a multi-role rental platform.
- Implemented tenant, owner, broker, and admin workflows, CSRF validation, role-based access checks, a custom PHP test suite, and MySQL-backed GitHub Actions CI.

RESEARCH EXPERIENCE

Undergraduate Researcher | Artificial Intelligence and Data Science Domain, AIUB 2024–Present

Explore machine-learning approaches for prediction and classification and NLP methods for conversational AI; support experimentation, documentation, and result analysis.

PROFESSIONAL EXPERIENCE

BD Chefs 2023–2025

Executive, Vendor Verification and Security Management | Dhaka, Bangladesh

- Reviewed vendor submissions, supported onboarding of home chefs and sellers, and maintained accurate verification records.
- Coordinated account-related issue resolution and supported vendor-security and verification procedures.

PUBLICATIONS & ACHIEVEMENTS

- **Publication:** *"A Large Language Model is Not the Right Path to Bring Artificial General Intelligence,"* 7th IEOM Bangladesh International Conference, Dec. 2024.
- **Academic Honors:** Dean's Award for Academic Excellence (Fall and Spring 2024–2025).
- **Awards:** 1st Place, 10th National Scientist Mania (2022; VirsysAI) | 1st Place, AI Solution Competition | 3rd Place, Web Design Competition (Notre Dame College, Mar. 2022).